

1. The Review Datasets Used and Their Original Structure

I. all_review.csv Dataset

- Rows: 12,000
- Columns: 11
- Important fields:
 - reviewText – the main review body
 - rating – numerical rating
 - reviewTime – review date (format: "MM DD, YYYY")
 - reviewerID, reviewerName, asin – identifiers for reviewer/book

II. book_review.csv Dataset

- Rows: 66,592
- Columns: 2
 - ratings – numeric (1 to 5)
 - reviews – freeform review text
- Additional review dataset loaded with utf-8 encoding and custom character handling.
- Used .applymap() to strip non-ASCII characters, indicating messy/unstructured text data.

III. Customer_Reviews.csv Dataset

- Rows: 910
- Columns: 11
 - reviewer, reviewer rating, review title, review description, date, ASIN, Author
 - book name – readable title.

2. Steps Taken for Standardization & Cleaning

I. Column Renaming & Selection

- Unified key fields across datasets:
 - Renamed columns to: review_id, user_id, book_id, review_text, rating, review_date, source
- Dropped unnecessary columns like Unnamed, summary, unixReviewTime, etc.

II. Date Standardization

- Converted all review dates to YYYY-MM-DD using pd.to_datetime()

III. Rating Normalization

- Verified all ratings are numeric (1–5)
- Removed or corrected inconsistent formats (e.g., text-based ratings)

IV. Text Cleaning

- Removed special characters/emojis.
- Removed non-ASCII characters and special characters.
- Cleaned and trimmed whitespace from review text fields

V. Encoding

- Standardized all text encoding to UTF-8 to avoid character corruption

VI. Source Tagging

- Added source column for each dataset:
 - "Amazon" → from all_review.csv
 - "Goodreads" → from book_review.csv
 - "Customer" → from Customer_Reviews.csv

3. Merging Strategy

- **Combined** all datasets into a single structured DataFrame with consistent schema using `pd.concat`.
- Removed duplicates using `merged_reviews.drop_duplicates(inplace=True)`
- Assigned a unique `review_id` to each row.

4. Issues Encountered and How They Were Resolved

I. Encoding Errors

- **Issue** found especially in `book_review.csv`
- **Solution:** Used utf-8 encoding with `engine='python'` and filtered out non-ASCII characters using `ord(x) < 128`.

II. Non-ASCII Characters in Review Text

- **Issue:** Review content had non-ASCII characters (e.g., emojis, accented characters) that could interfere with text processing.
- **Resolution:**
Applied a lambda function using `applymap()` to clean all string fields

III. Inconsistent Column Names Across Datasets

- **Issue:** Review datasets had different column names for similar content (e.g., `review`, `text`, `comment`, or `stars`, `rating`).
- **Resolution:**
Although not shown in the code, it's implied that column names were or should be renamed and standardized before merging (e.g., `review`, `rating`, `book_id`).